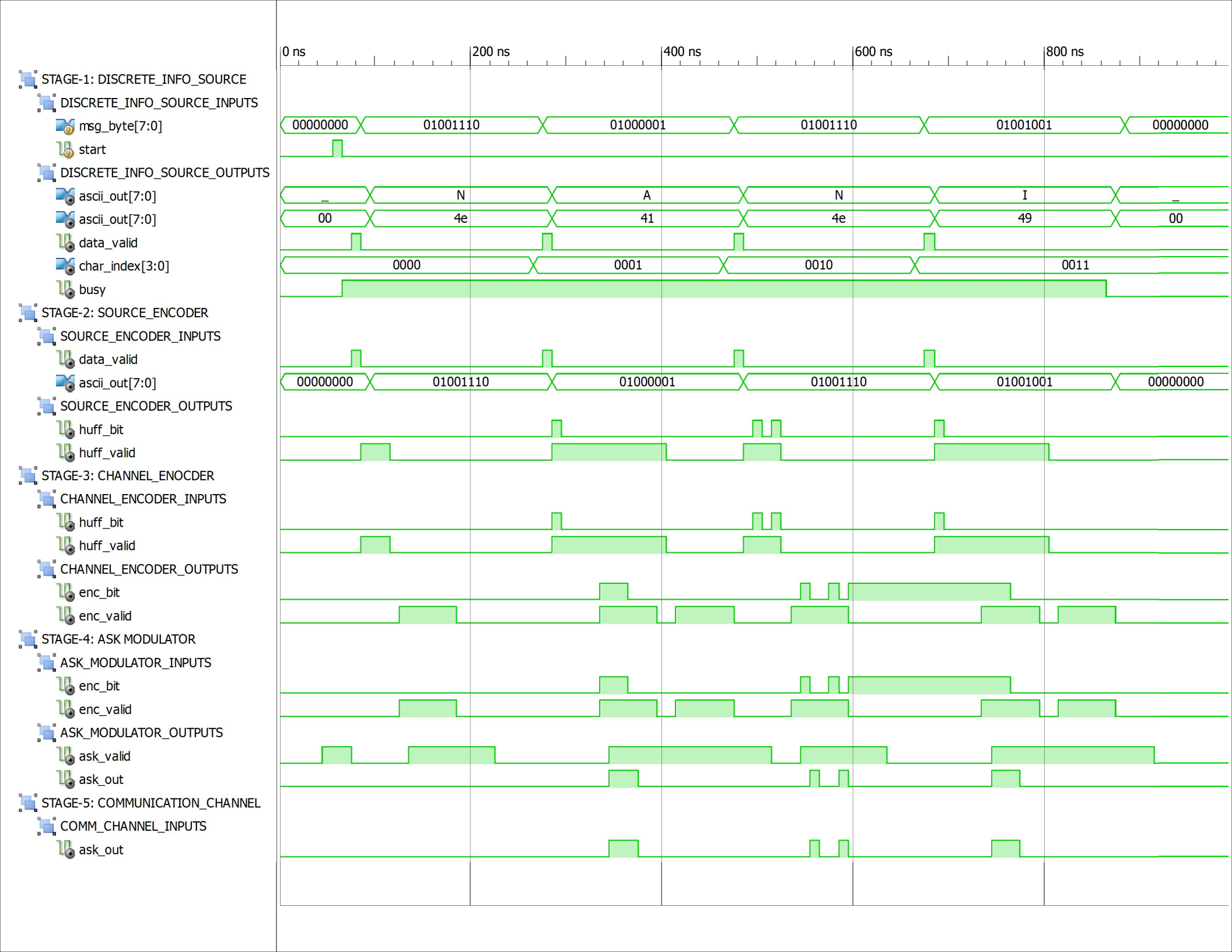
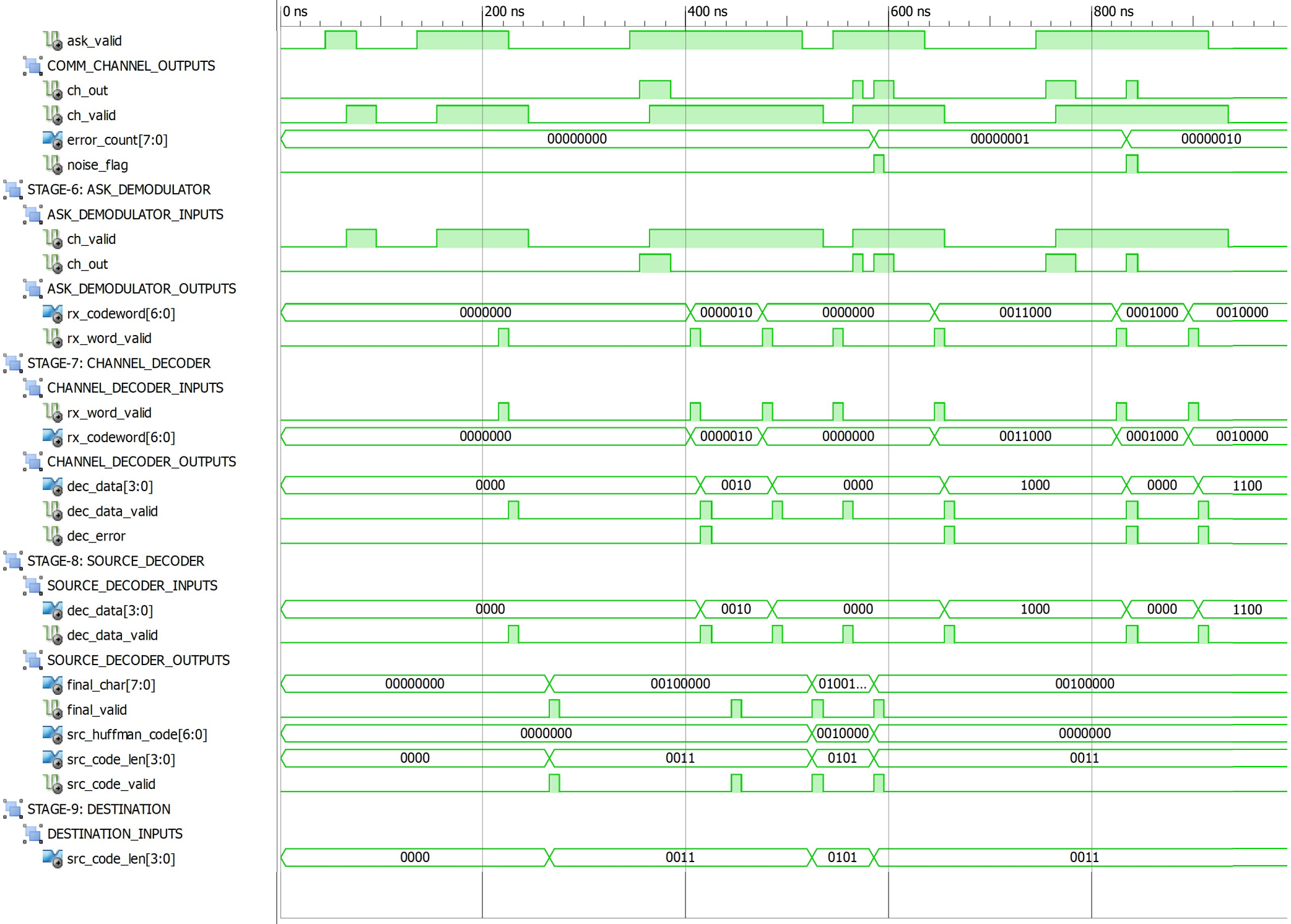
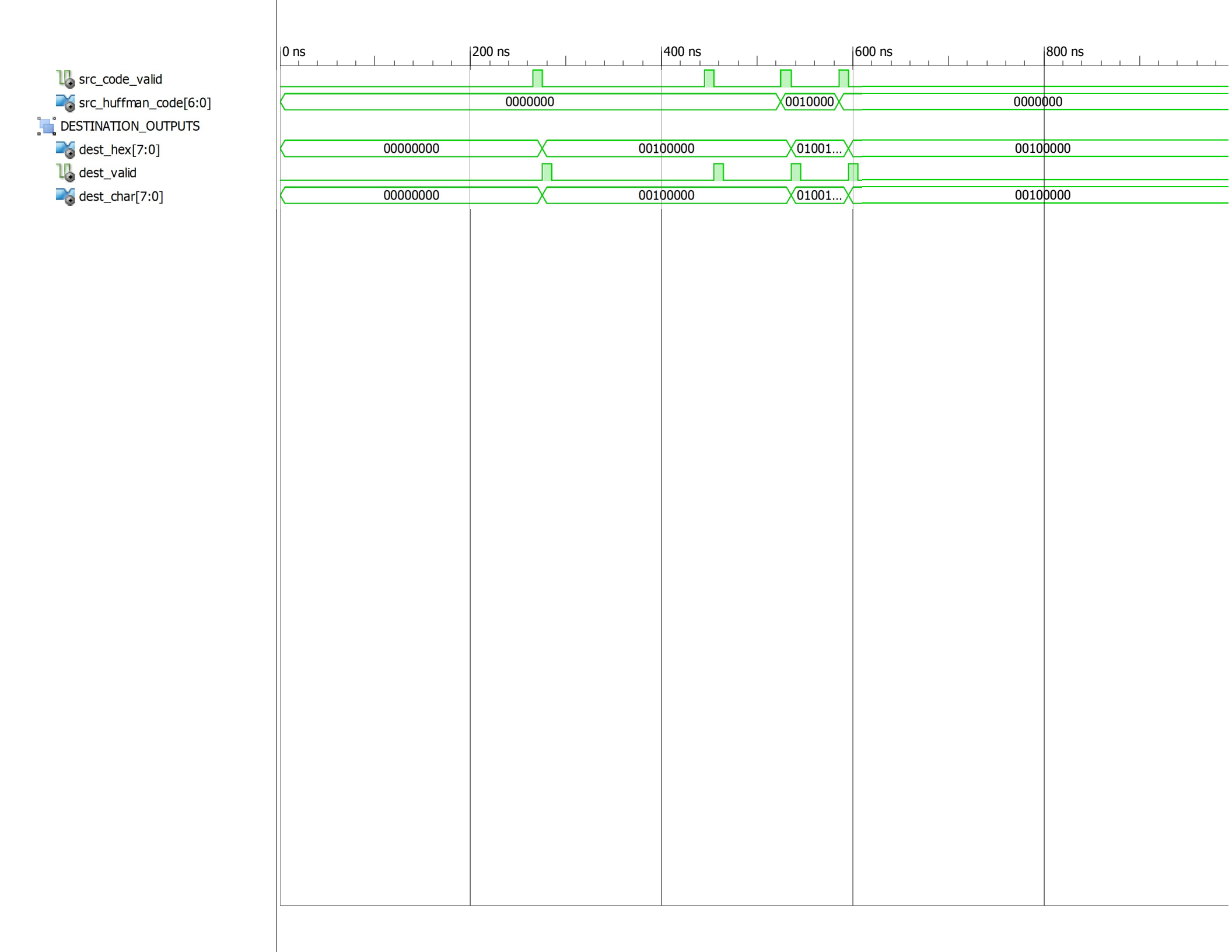

0 ns
200 ns
400 ns
600 ns
800 ns
STAGE-1: DISCRETE_INFO_SOURCE
DISCRETE_INFO_SOURCE_INPUTS
msg_byte[7:0]
start
DISCRETE_INFO_SOURCE_OUTPUTS
ascii_out[7:0]
ascii_out[7:0]
data_valid
char_index[3:0]
busy
STAGE-2: SOURCE_ENCODER
SOURCE_ENCODER_INPUTS
data_valid
ascii_out[7:0]
SOURCE_ENCODER_OUTPUTS
huff_bit
huff_valid
STAGE-3: CHANNEL_ENOCDER
CHANNEL_ENCODER_INPUTS
huff_bit
huff_valid
CHANNEL_ENCODER_OUTPUTS
enc_bit
enc_valid
STAGE-4: ASK MODULATOR
ASK_MODULATOR_INPUTS
enc_bit
enc_valid
ASK_MODULATOR_OUTPUTS
ask_valid
ask_out
STAGE-5: COMMUNICATION_CHANNEL
COMM_CHANNEL_INPUTS
ask_out
00000000
01001110
01000001
01001110
01001001
00000000
_
N
A
N
I
_
00
4e
41
4e
49
00
0000
0001
0010
0011
00000000
01001110
01000001
01001110
01001001
00000000


0 ns
200 ns
400 ns
600 ns
800 ns
ask_valid
COMM_CHANNEL_OUTPUTS
ch_out
ch_valid
error_count[7:0]
noise_flag
STAGE-6: ASK_DEMODULATOR
ASK_DEMODULATOR_INPUTS
ch_valid
ch_out
ASK_DEMODULATOR_OUTPUTS
rx_codeword[6:0]
rx_word_valid
STAGE-7: CHANNEL_DECODER
CHANNEL_DECODER_INPUTS
rx_word_valid
rx_codeword[6:0]
CHANNEL_DECODER_OUTPUTS
dec_data[3:0]
dec_data_valid
dec_error
STAGE-8: SOURCE_DECODER
SOURCE_DECODER_INPUTS
dec_data[3:0]
dec_data_valid
SOURCE_DECODER_OUTPUTS
final_char[7:0]
final_valid
src_huffman_code[6:0]
src_code_len[3:0]
src_code_valid
STAGE-9: DESTINATION
DESTINATION_INPUTS
src_code_len[3:0]
00000000
00000001
00000010
0000000
0000010
0000000
0011000
0001000
0010000
0000
0010
0000
1000
0000
1100
00000000
00100000
01001...
00100000
0000000
0010000
0000000
0000
0011
0101
0011


0 ns
200 ns
400 ns
600 ns
800 ns
src_code_valid
src_huffman_code[6:0]
DESTINATION_OUTPUTS
dest_hex[7:0]
dest_valid
dest_char[7:0]
0000000
0010000
0000000
00000000
00100000
01001...
00100000
00000000
00100000
01001...
00100000